TP2: Launch and Secure a Cloud Virtual Machine (IaaS)

1. Objectives

- Understand the Infrastructure as a Service (IaaS) model and the difference between a local VM and a cloud VM.
- Launch a small Linux virtual machine on an approved public cloud platform (AWS, Microsoft Azure, or Google Cloud).
- Configure network access using the principle of least privilege.
- Connect to the VM remotely using SSH and verify the remote host.
- Install and test the Nginx web server.
- Identify cloud characteristics observed during the lab and estimate the resources/costs used.
- Stop or delete resources safely and verify cleanup.

2. Scenario

You are a junior DevOps engineer asked to deploy a simple website on a cloud server. You must provision a Linux VM, secure remote access, install a web server, test it from a browser, and clean up the environment when finished.

Part A: Choose the platform and create a Linux VM

1. Sign in to one provider: AWS, Microsoft Azure, or Google Cloud. Use only one provider for this lab.
2. Open the VM/compute service (for example, EC2 on AWS, Virtual Machines on Azure, or Compute Engine on Google Cloud).
3. Choose a small Linux image, such as Ubuntu LTS, and a free-tier-eligible or instructor-approved size. Choose a nearby region.
4. Use a clear resource name, for example: tp2-<student-initials>. Do not include passwords or personal sensitive data in names.
5. Create or select an SSH key pair. Download the private key only to your own computer and store it securely. Never share it.
6. Configure the network/firewall rules before launch: SSH/TCP 22 from your public IP only; HTTP/TCP 80 from your IP for testing (or follow the instructor's rule). Do not open all ports.
7. Review the summary, including region, VM size, storage, public IP option, and any estimated cost. Launch the VM.
8. Wait until the provider reports that the VM is running and its health/status checks pass. Record the region, image, size, and public IP.

Part B: Connect to the VM using SSH

Use the public IP shown in the cloud console. The default username depends on the image/provider (for Ubuntu, it is often ubuntu, but verify the provider's instructions).

Linux/macOS or Windows Terminal with OpenSSH:

```
ssh -i /path/to/private-key.pem ubuntu@<PUBLIC_IP>
```

On Windows, the key path may look like C:\Users\YourName\Downloads\mykey.pem. Some providers require specific key permissions. Follow the provider's instructions if SSH rejects the key.

After connecting, run:

- `hostname`
- `whoami`
- `ip a`
- `uptime`

If the connection fails, check: the VM is running; the public IP is correct; SSH is allowed from your current IP; the username is correct; and the private key matches the VM. Never solve this by opening SSH to the whole Internet.

Part C: Install and test Nginx

Inside the cloud VM, execute:

- `sudo apt update`
- `sudo apt install -y nginx`
- `sudo systemctl enable --now nginx`
- `sudo systemctl status nginx --no-pager`

From your own computer, open this URL in a browser: http://<PUBLIC_IP>. You should see the Nginx welcome page. If it does not load, verify that TCP port 80 is allowed by the cloud firewall/security group and by the VM's operating-system firewall, if enabled. Do not disable the firewall entirely.

Optional check from the VM:

```
curl -I http://localhost
```

Part D: Observe cloud characteristics

Answer using evidence from your own deployment. Do not just copy definitions.

1. On-demand self-service: Which action did you perform yourself in the cloud console?
2. Broad network access: How did you access the VM or its website over a network?
3. Resource pooling: Which provider-managed resources are abstracted from you?
4. Rapid elasticity: Could the VM size be changed? What would you need to check first?
5. Measured service: Where can you inspect usage, billing, or cost estimates? If you had no billing access, state that clearly.

Part E: Clean up resources

1. Save the required screenshots and notes first, without exposing secrets.
2. Use the provider console to terminate/delete the VM (or follow the instructor's explicit instruction if a shared demonstration VM is being used).

3. Check whether disks, snapshots, static/reserved IP addresses, load balancers, or other created resources remain. Delete only resources created for this lab and no longer needed.
4. Verify that the VM is no longer running and record the final state.

3. Questions

Q10. Which cloud provider did you use? State the region, Linux image, VM size, and public IP (the public IP may be masked in a public submission if required by the instructor).

Answer:
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Q2. Explain two differences between the VMware VM from TP1 and this cloud VM.

Answer:
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Q3. What is IaaS? Identify which infrastructure components are managed by the provider and which tasks remain your responsibility.

Answer:
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Q4. Why should SSH access be restricted to your IP address? What could go wrong if TCP port 22 were open to everyone?

Answer:
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Q5. What evidence confirms that your SSH session is connected to the cloud VM rather than your local computer?

Answer:
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Q6. What did you do to verify that Nginx was running and reachable from your browser?

Answer:
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Q7. Name two of the five essential cloud characteristics and explain how you observed each one.

Answer:
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Q8. Did you find a cost estimate or usage page? State what you checked. If unavailable, explain why you could not verify it.

Answer:
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Q9. Which resources did you delete? Why is stopping the VM not always enough to avoid charges?

Answer:
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4. Submit

1. Screenshot 1: cloud console showing the VM running, with its region and size visible. Hide account IDs and other sensitive information.;
2. Screenshot 2: terminal showing the SSH session and the outputs of hostname and whoami. Do not show private keys or credentials.
3. Screenshot 3: Nginx welcome page loaded from http://<PUBLIC_IP>.
4. Screenshot 4: provider console confirming that the VM is terminated/deleted and no lab resources remain running.
5. Short report (about 1–2 pages) containing answers to Q1–Q9